

MTI SMART PARKING SOLUTION

Misr Trade & Investment (MTI)

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Technical-Commercial Proposal Study

For

نايف الراجحي الاستثمارية
NAIF ALRAJHI INVESTMENT



حديقة الملك سلمان
King Salman Park



NRI District Smart City

- Indoor Basement Parking Implementation for 31 Building

Scope

- Smart Parking Management &
- Smart Parking Guidance Management System

Technology

- Ultrasonic Sensors for Parking Guidance
- Custom made Smart Parking Management & Guidance Software



About Our Company

Misr Trade & Investment (MTI) represents a group of organizations with diversified fields of operations, working under the supervision of highly experienced and reputed management. The diversification of this group is the consequence of visionary leadership and determination to progress.

Our Vision

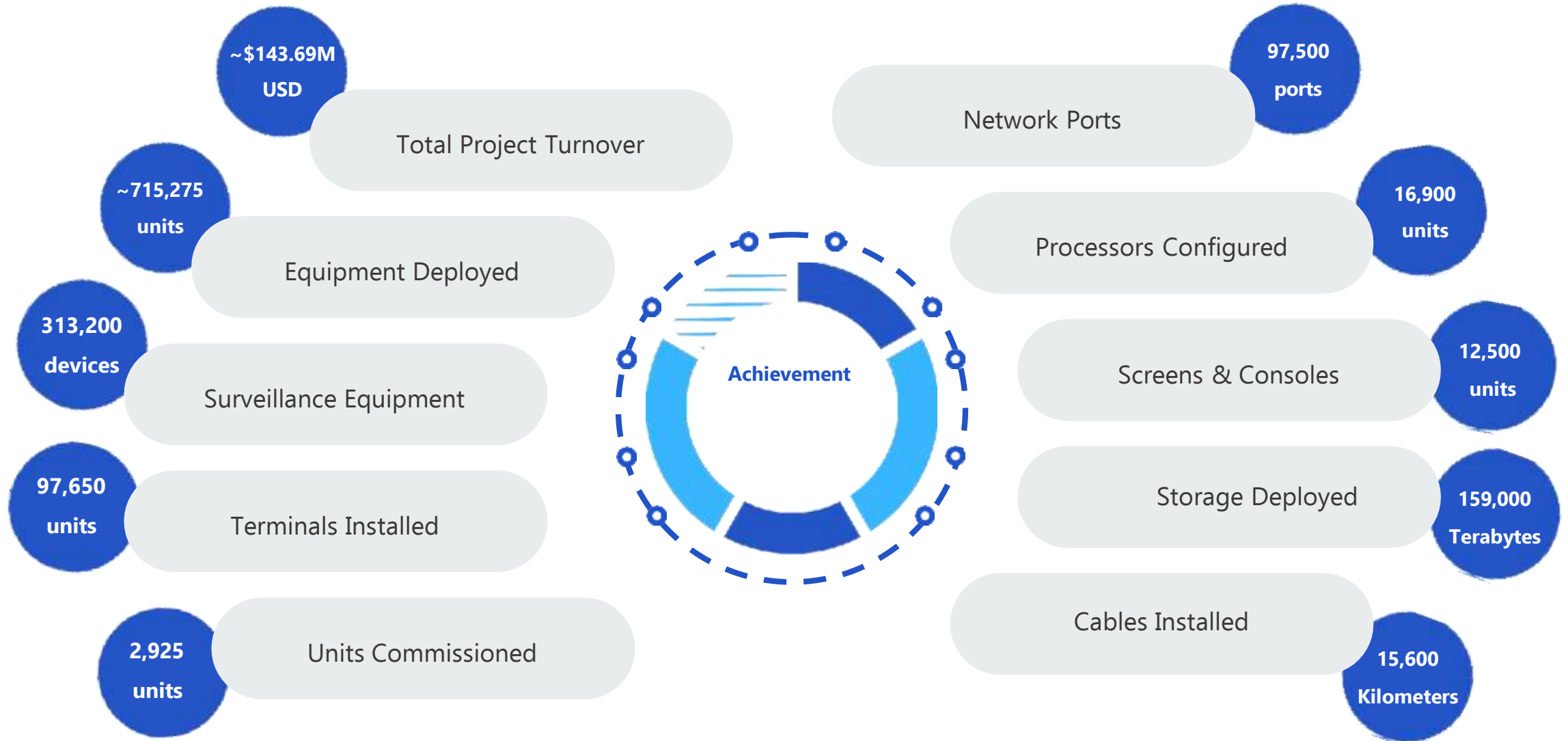
Empower people by bringing inspiration & innovation to their life enabling them to live better & easily.

Our Mission

To be the #1 company in the MENA region during the upcoming 5 years by providing latest technology solutions serving humanity in different aspects in order to facilitate their life with continuous development.

5-Year Track Record

MTI has demonstrated exceptional capability in delivering large-scale technology projects across the MENA region:



EXECUTIVE SUMMARY

This document presents a comprehensive Smart Parking Solution—a fully integrated, intelligent parking management platform designed to deliver seamless experiences for residents, employees, and visitors across the NRI district.

The solution combines proven Parking Management system with advanced parking guidance Ultrasonic technology, offering flexible deployment options that balance cost-effectiveness with cutting-edge innovation.

This platform ensures efficiency, automation, and exceptional user convenience.





Key Highlights

- **Scalable Architecture**
Modular four-layer design adaptable from a single facility to multi-site or community-wide deployments.
- **System Reliability**
Runs continuously with backup layers to prevent downtime, protect revenue, and keep data safe. Easily scales to any number of sites and is built to support future growth
- **Technology**
Supports ultrasonic sensor technology for accurate, real-time parking occupancy detection.
- **Special User Access**
Dedicated parking slots equipped with ANPR (Automatic Number Plate Recognition) cameras and automated slot blocker barriers for reserved or VIP users.
- **Multi-Zone Management**
Ensures guaranteed parking within primary zones while enabling seamless access to secondary or outdoor parking areas.
- **Smart City Integration**
Natively connects with urban mobility and smart city platforms for centralized management and data-driven insights

Project Scope

- **Deployment Environment**
Indoor basement parking (underground)
across 31 buildings
- **System Type**
Parking management system (PMS) +
Ultra sonic Parking Guidance system
(UPGS)
- **Technology**
Ultrasonic Sensors.
- **Access Control**
Controlled entry/exit gates Scalability:
Solution replicable across all 31 buildings
with economies of scale

Key Project Highlights

Aspect	Details
Facility Scope	6000 slots, 2 entry & 1 exit lanes in each building
Architecture	4-layer design for scalability and integration
Technology Options	Ultrasonic Sensor
Smart City Integration	Smart city Platforms connectivity

Smart Parking Management System - Complete Solution Overview

1. System Architecture

The Smart Parking Management System provides comprehensive parking management across 31 buildings through a three-tier architecture designed for high availability, scalability, and operational continuity.

Central Management System

Primary Function: Enterprise-wide control and monitoring of all parking facilities

Core Capabilities:

- Real-time monitoring of all 31 buildings simultaneously
- Consolidated reporting and business intelligence
- Revenue management and financial reconciliation
- User database management (citizens, employees, visitors)
- Subscription bundle creation and management
- System-wide configuration and policy control
- High Availability (HA) dual-server configuration for 99.99% uptime
- Automatic failover protection

Key Features:

- Unified dashboard displaying occupancy status across all facilities
- Cross-facility analytics and performance metrics
- Centralized user management and access control
- Multi-building subscription management
- Comprehensive audit trails and compliance reporting

Building-Level Management

Primary Function: Independent operation and management of individual parking facilities

Operator Access: Each building has dedicated operators with building-specific permissions to:

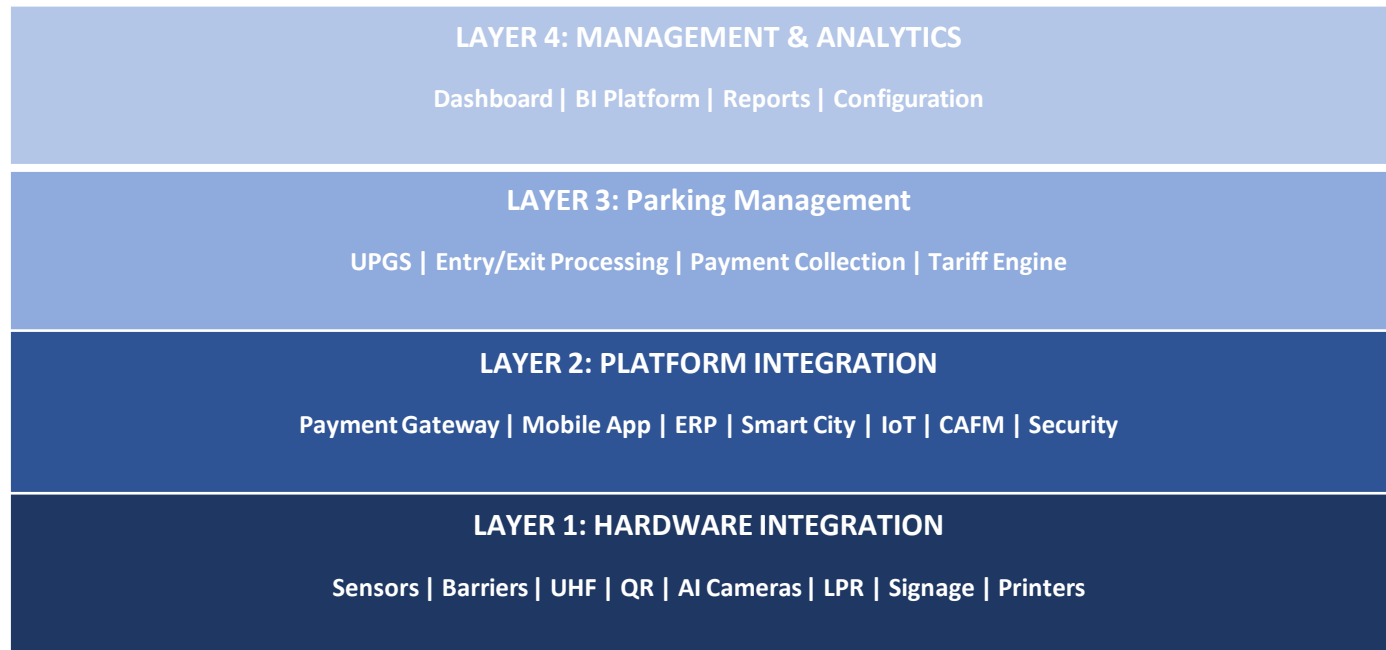
- Monitor real-time parking occupancy for their facility only
- Manage local subscriptions and reservations
- Search for vehicles by license plate
- Generate and assign UHF tags
- Process manual transactions and override requests
- View building-specific reports and analytics
- Control entry/exit gates and hardware devices

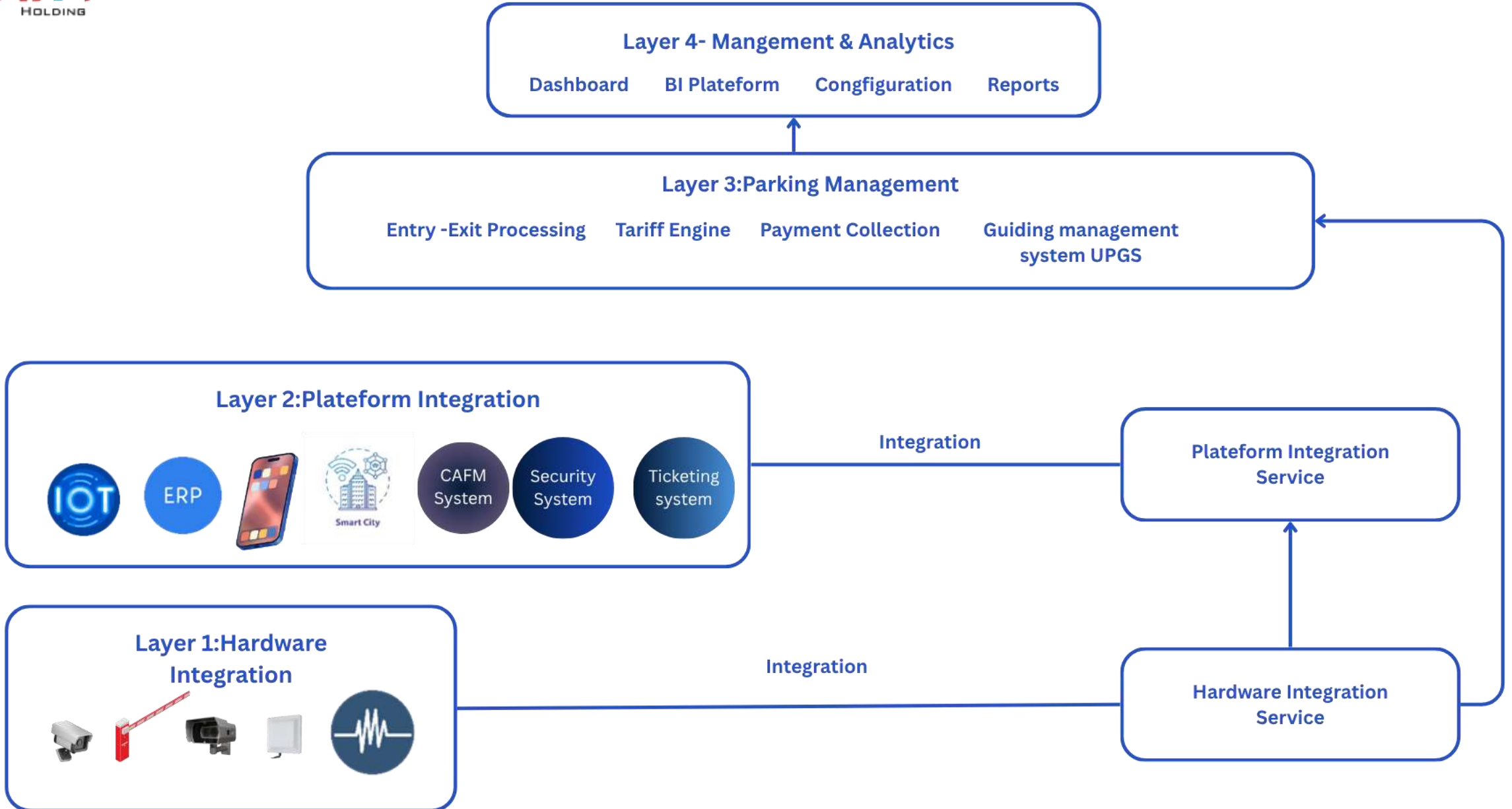
Independent Operation:

- Each building operates autonomously with dedicated server
- Local database ensures continuity during network interruptions
- Hardware integration managed at building level
- Automatic synchronization with central system when connected

Layer Architecture Model

The system employs a modern 4-layer architecture designed for optimal scalability, maintainability, and integration capabilities

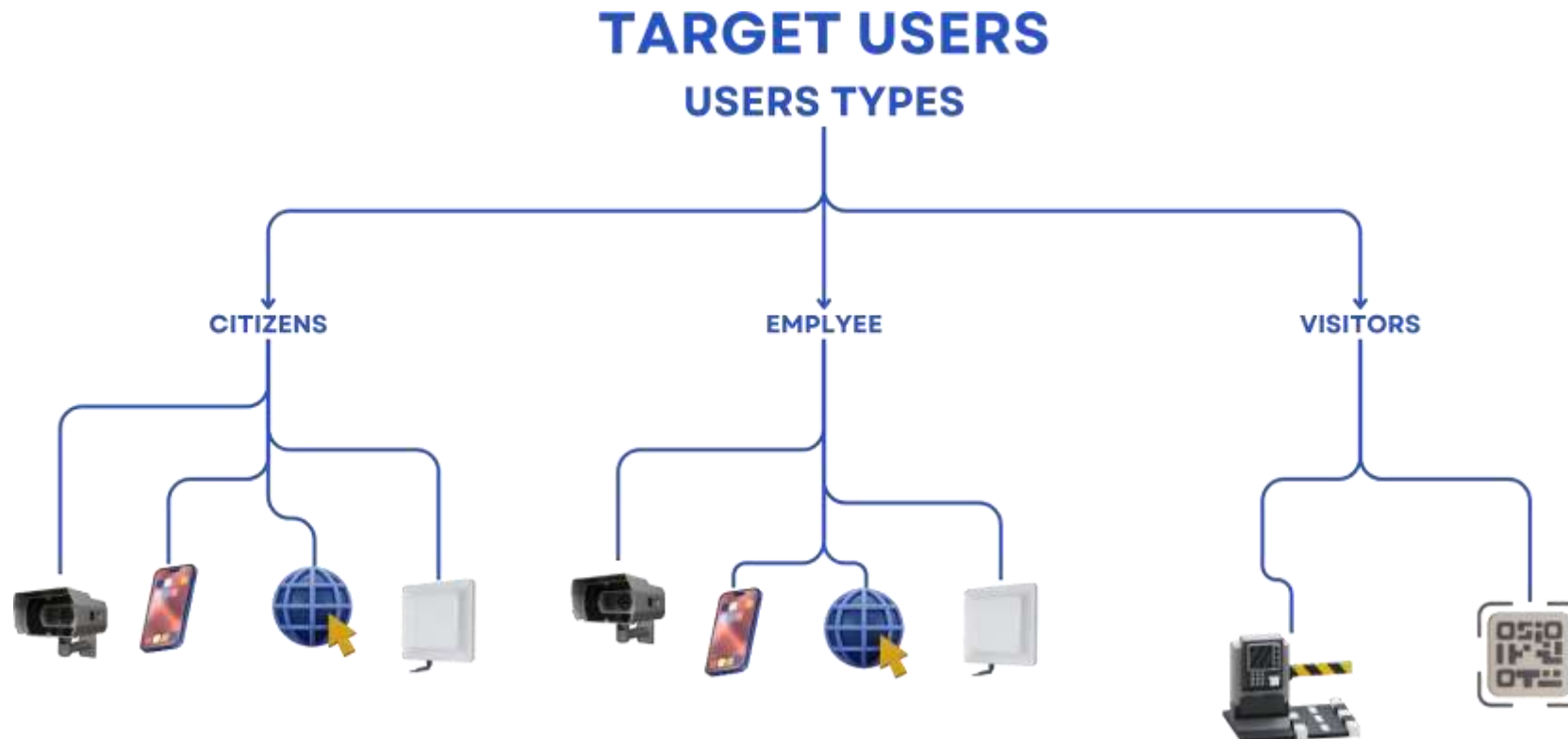




2. USER MANAGEMENT & ACCESS CONTROL

The system supports multiple entry and exit methods to accommodate different user types and scenarios:

- 1 UHF Subscription Access
- 2 QR Code & LPR Access
- 3 Manual Entry (Visitor/Guest) (Ticketless As Option)
- 4 Ticketless Entry Option



Residents

Regular users who can subscribe for parking access across designated areas.

Access Methods:

- UHF tag (contactless card)
- QR Code + LPR combination
- Mobile app with digital credentials

Subscription Channels:

- Citizen mobile application
- City web portal
- Public KIOSK terminals

Employees

Organizational staff with dedicated parking privileges.

Features:

- Organization-specific parking zones
- Configurable access rights by department
- Integration with HR/employee management systems

Visitor

Temporary users requiring short-term parking access.

Access Options:

- Invited by citizens or employees
- Reserved parking for defined duration
- Manual ticket issuance via push button
- QR Code and LPR validation

Payment Methods:

- KIOSK payment stations (Digital Payment)
- POS cashier terminals (Cash-Digital Payment)
- Mobile app pre-payment
- Credit/debit cards at exit POS (Optional Add POS For Payment On Exit Gateway)
- City Web Platform

USER MANAGEMENT & ACCESS CONTROL

Subscription Model

The Smart Parking System allows flexible subscription management across multiple parking areas:

Each resident subscription includes:

- One main assigned parking area, where a dedicated slot is reserved for the full subscription duration.
- Optional temporary parking areas (e.g., malls, school, or public facilities):
 - The subscriber can access these optional parks if a slot is available at arrival.
 - No slot reservation is guaranteed for these optional areas.
 - If all slots are occupied, the subscriber will not have a pre-reserved space in that location.

Example:

An administration may create a special subscription bundle for specific citizens.

The bundle includes a guaranteed reserved slot in the main administrative parking and access rights to optional city parking zones.

The subscriber can park freely in optional zones if space is available.

USER MANAGEMENT & ACCESS CONTROL

Subscription Model

Each Employees subscription includes:

- Have the same subscription and access options as residents.
- Employee subscriptions can be configured for specific organizational or administrative parking zones.

Each Visitor subscription includes:

- Can be invited by residents or employees.
- May reserve parking slots for a defined duration.
- Access methods:
 - QR Code and LPR
 - Manual ticket issuance via push button
- Payment options:
 - KIOSK stations
 - POS cashier terminals

USER MANAGEMENT & ACCESS CONTROL

Parking System – Entry & Exit Scenarios

UHF Subscription Entry

1. System validates UHF tag.
2. If valid → Confirm subscriber has pre-reserved slot in main parking area.
3. Gate opens automatically → Operator notified with subscriber details.
4. Transaction recorded (LPR image, subscriber info, vehicle details, gate ID).
5. Digital signage guides driver to designated slot.
6. Slot LED indicator turns red (occupied).
7. For optional areas: Subscriber may enter if subscription valid and free slot available.

QR Code & LPR Subscription / Reservation Entry

1. User scans QR code at entrance.
2. System performs: LPR recognition → QR–LPR data validation → Slot availability check.
3. If validation succeeds → Barrier opens automatically → Transaction recorded.
4. Digital signage guides user to available slot.
5. Slot LED indicator turns red when occupied.
6. If validation fails → User presses manual button to obtain printed ticket and enter manually.

Manual Entry

1. User presses push button to issue manual entry ticket.
2. System captures LPR data → Opens barrier → Records entry event.
3. All entry data synchronized automatically with: Main Smart City Platform & IoT Platform.

USER MANAGEMENT & ACCESS CONTROL

Parking System – Entry & Exit Scenarios

UHF Subscription Exit

1. System validates UHF tag at exit.
2. If valid → Gate opens automatically → Operator notified with subscriber details.
3. Exit transaction recorded (LPR image, subscriber info, gate ID).

QR / LPR Subscription, Reservation, or Manual Ticket Exit

1. User scans QR code at exit.
2. System checks: QR/LPR exit authorization or manual ticket payment status.
3. If payment required → complete payment at KIOSK or POS cashier.
4. after payment → 20-minute grace period (customizable) to exit.
5. Supported payment methods: Cash, Visa/Credit Card, Digital Wallet, other integrated gateways.

For Citizens and Employees:

- Renew existing subscriptions
- Purchase new subscription bundles
- Make parking reservations
- View parking history
- Update vehicle information
- Print receipts and confirmations

System Integration:

- Connected to building server
- Real-time validation against central database
- Transaction logging and synchronization
- Automatic grace period activation

2 POS Cashier System

Location: Cashier station in each building [All Kiosk Functions PLUS:](#)

- Cash payment acceptance (bills and coins)
- Manual transaction processing
- Lost ticket resolution
- Dispute handling and adjustments
- Receipt printing for all payment types
- Operator-assisted services

[Enhanced Capabilities:](#)

- Grace period overrides
 - Special rate applications
 - Manual voucher redemption
 - Complex transaction handling
-

3. Parking Guidance System (UPGS Integration)

- 1 Standard Parking Spaces
- 2 Reserved Parking Spaces

4. Payment Infrastructure :

1 Digital Kiosk System

- Location: One kiosk installed in each building basement

User Functions:

For Visitors:

- Scan parking ticket
- View parking duration and fee calculation
- Process payment via:
 - Credit/debit cards
 - UHF prepaid cards
 - Mobile payment (digital wallet)
- Print receipt with validated ticket
- 20-minute grace period for exit

5. Grace Period Management

Default Settings:

- Standard grace period: 20 minutes (configurable 15-30 minutes)
- Starts from payment completion timestamp
- Allows reasonable time to locate vehicle and exit

Enforcement:

- System prevents exit if grace period exceeded
- Additional fee calculated automatically (e.g., \$2 per 15 minutes)
- Driver directed to cashier for supplemental payment
- Logged as "grace period violation" for analytics

Special Considerations:

- Disabled parking: Extended grace period (30-45 minutes)
- VIP users: No grace period enforcement
- Emergency situations: Manual operator override

6. System Resilience and Failover

1. Normal Operations
2. Building Server Failure
3. Centralized Server Communication Loss
4. Complete Communication Loss

7. Data Synchronization

Real-Time Integration

Smart City Platform:

- Live occupancy updates
- Transaction events
- Parking duration statistics
- Revenue data

IoT Platform:

- Sensor status updates
- Equipment health metrics
- Environmental data
- Performance analytics

ERP System:

- Financial transactions
- Revenue accounting
- Cash flow tracking
- Reconciliation data

Mobile Application:

- User parking history
- Current parking session status
- Payment receipts
- Reservation confirmations

Offline Operation:

- Local database caching at building level
- Transaction queuing during network outage
- Automatic synchronization when connection restored
- Manual operations continue via local controllers
- Data integrity maintained throughout

8. Administrative Functions

1. User Management

Central Administration:

- Create and manage user accounts (citizens, employees, visitors)
- Assign parking privileges and access rights
- Define subscription tiers and pricing
- Configure payment methods and rates
- Whitelist management for VIP and special users

Building-Level Administration:

- View and search users assigned to their facility
- Process subscription renewals
- Generate UHF tags
- Handle reservation requests
- Resolve customer issues

2. Subscription Management

Bundle Creation:

- Define subscription packages (monthly, quarterly, annual)
- Set pricing structures
- Specify included parking areas (main/optional)
- Configure slot reservation privileges
- Apply special rates (employee discounts, bulk rates)

Tag Generation:

- Create and encode UHF tags
- Associate tags with user accounts and vehicles
- Deactivate lost or stolen tags
- Batch tag generation for corporate accounts

3. Reservation System

Capabilities:

- Time-based slot reservations
- Recurring reservation scheduling
- Conflict detection and prevention
- Cancellation and modification handling
- Waitlist management for high-demand slots

4. Vehicle Search

Search Functions:

- Locate vehicle by license plate across all 31 buildings
- View current parking location and duration
- Display entry timestamp
- Show parking history for registered vehicles
- Support for partial plate searches

Use Cases:

- Security investigations
 - Enforcement verification
 - Customer service support
-

9. Analytics and Reporting

1. Real-Time Dashboard

System-Wide Metrics:

- Total occupancy across all facilities
- Available spaces per building
- Revenue generation (hourly/daily/monthly)
- Active subscriptions count
- Current reservations
- Building-Level Metrics:
 - Facility-specific occupancy rates
 - Peak usage times
 - Average parking duration
 - Local revenue performance
 - Equipment status alerts

2. Standard Reports

Operational Reports:

- Daily transaction summary
- Entry/exit activity logs
- Payment collection report
- Grace period violations
- Lost ticket incidents
- Financial Reports:
 - Revenue by payment method
 - Subscription revenue vs. visitor revenue
 - Cashier reconciliation
 - Outstanding balances
- Refund/adjustment tracking
- Performance Reports:
 - Occupancy trends and patterns
 - Space utilization efficiency
 - Peak demand analysis
 - User behavior analytics
 - Hardware uptime statistics

10. Scalability and Future Expansion

Horizontal Scalability

- Add unlimited additional buildings to the system
- Each facility operates independently
- Central management scales automatically

Vertical Scalability

- Increase capacity per building as needed
- Add more entry/exit lanes
- Expand parking space monitoring
- Integrate additional payment terminals

Modular Design

- Add new features without disrupting operations
 - Integrate with future smart city initiatives
 - Support for emerging payment technologies
 - Ready for autonomous vehicle integration
-

11. System Benefits Summary

Capability	Benefit
Multi-Building Management	Unified control of 31 facilities from single dashboard
High Availability	99.99% uptime with automatic failover protection
Revenue Protection	Exit operations guaranteed even during system failures
Operational Flexibility	Multiple access methods (UHF, QR, LPR, manual)
User Convenience	Pre-payment kiosks, mobile app, ticketless options
Real-Time Guidance	LED indicators and digital signage reduce search time
Special Slot Enforcement	LPR-based automated monitoring of VIP, disabled, EV, reserved spaces
Data Integrity	Zero transaction loss with automatic synchronization
Scalability	Support unlimited facilities with consistent performance
Comprehensive Reporting	Real-time analytics and customizable reports
Security	End-to-end encryption, audit trails, violation detection

Parking Management System - High-Level Architecture Overview

1. Architecture Components

1.Three-Tier Architecture Centralized Management Tier

- High Availability (HA) dual-server configuration
- Enterprise-wide data consolidation and reporting
- Failover management and business intelligence
- 24/7 monitoring and control

Building Facility Tier

- Dedicated PMS/UPGS server per building
- Local parking and guidance management
- Integration with all hardware devices
- Independent operation capability

Hardware & Payment Tier

- Entry/exit barriers and vehicle sensors
- Parking guidance sensors and displays
- Point-of-Sale (POS) payment terminals
- License plate recognition cameras

2. Key System Capabilities

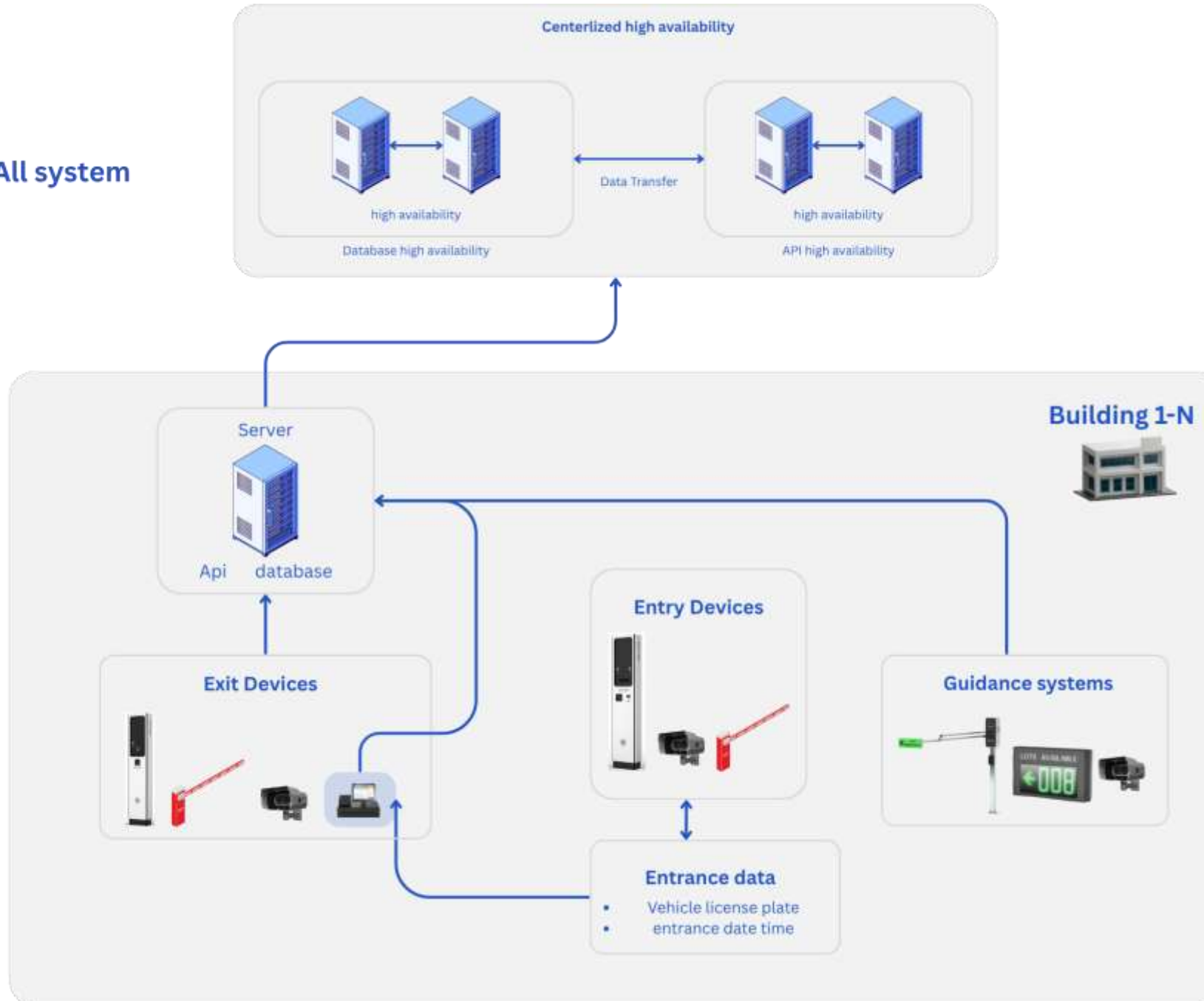
1. Normal Operations All Systems Online:

- Building servers manage local parking operations
- POS terminals process payments through building servers
- Real-time data synchronization to centralized servers
- Complete visibility and control across all facilities
- Entry and exit operations fully functional

Data Flow and Synchronization

- Hardware Devices → Building Server → Centralized Servers
 - POS Terminals → Building Server → Centralized Servers
 - Entrance Data → POS Terminals
-

Normal Operation All system Online



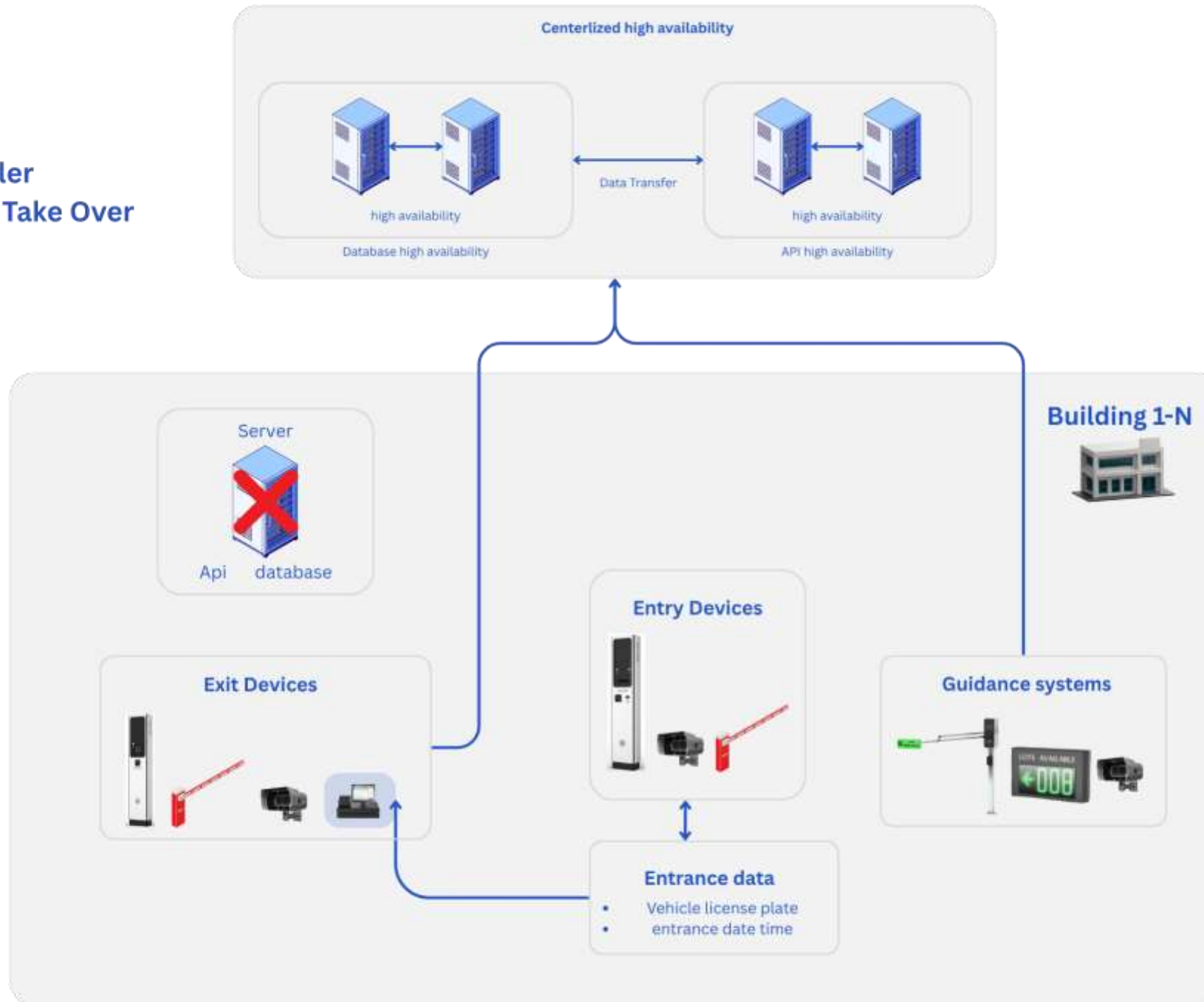
2. Building Server Failure Centralized Servers Take Over:

- Automatic failover detected within seconds
- Centralized servers assume parking management control
- POS terminals connect directly to centralized servers
- Hardware control transferred seamlessly
- Entry and exit operations continue without interruption
- All transaction data preserved and synchronized upon recovery

Data Flow and Synchronization

- Hardware Devices → Centralized Servers
 - POS Terminals → Centralized Servers → Building Server (upon recovery)
 - Entrance Data → POS Terminals
-

Building server frailer Centralized Center Take Over



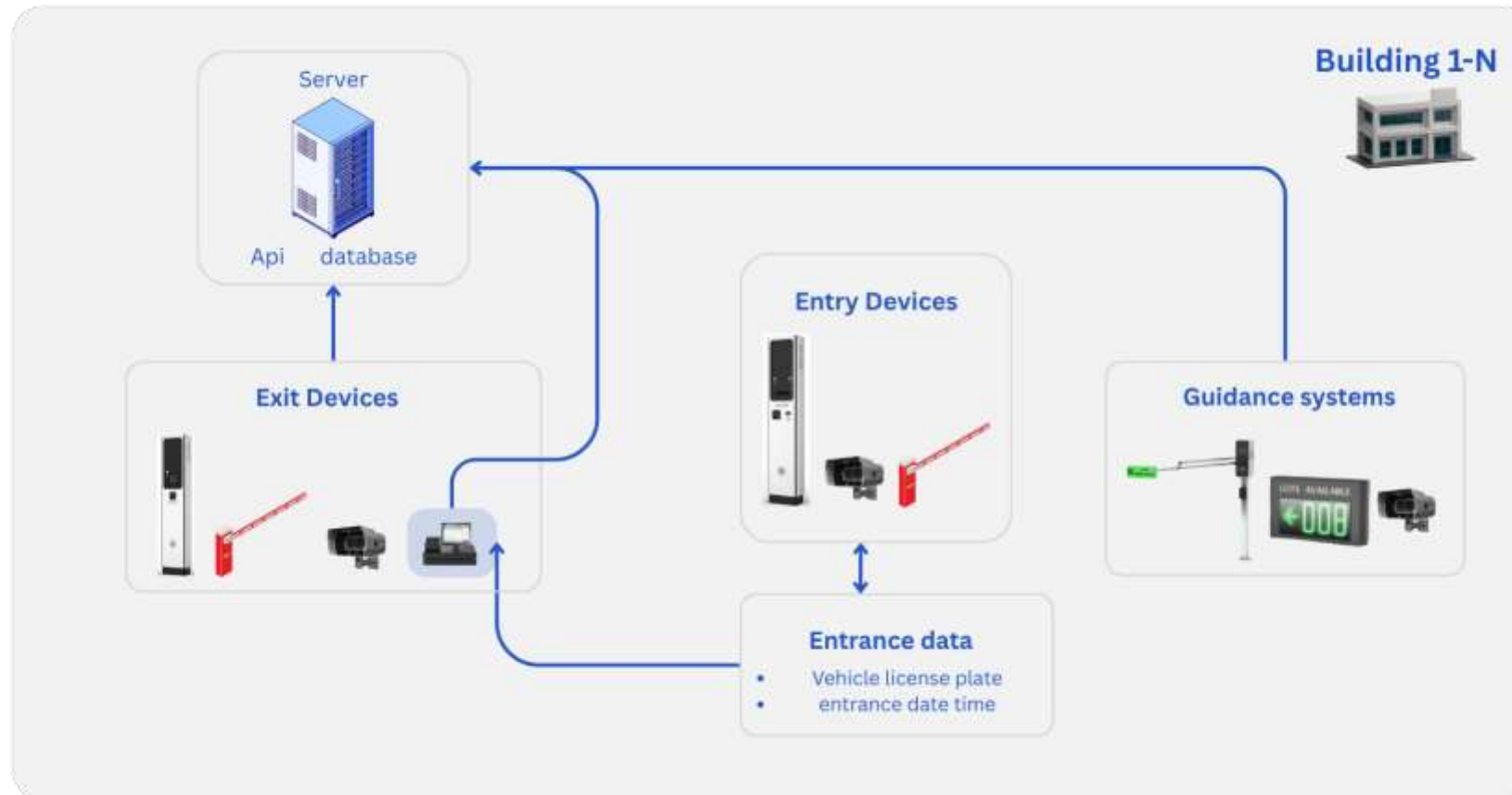
2.3 Centralized Server Communication Loss

- Building Server Continues Normal Operations:
- Building servers operate independently without centralized connectivity
- POS terminals continue processing payments through building servers
- Entry and exit operations fully functional
- All transactions stored locally on building server database
- Real-time parking management and guidance systems unaffected
- Data queued for synchronization to centralized servers
- Upon centralized server connectivity restoration, automatic data replication occurs
- Zero operational impact on parking facility operations
- Centralized reporting and cross-facility visibility temporarily unavailable

Data Flow and Synchronization

- Hardware Devices → Building Server (upon recovery)
- POS Terminals → Building Server (upon recovery)
- Entrance Data → POS Terminals

Centralized server communication loss



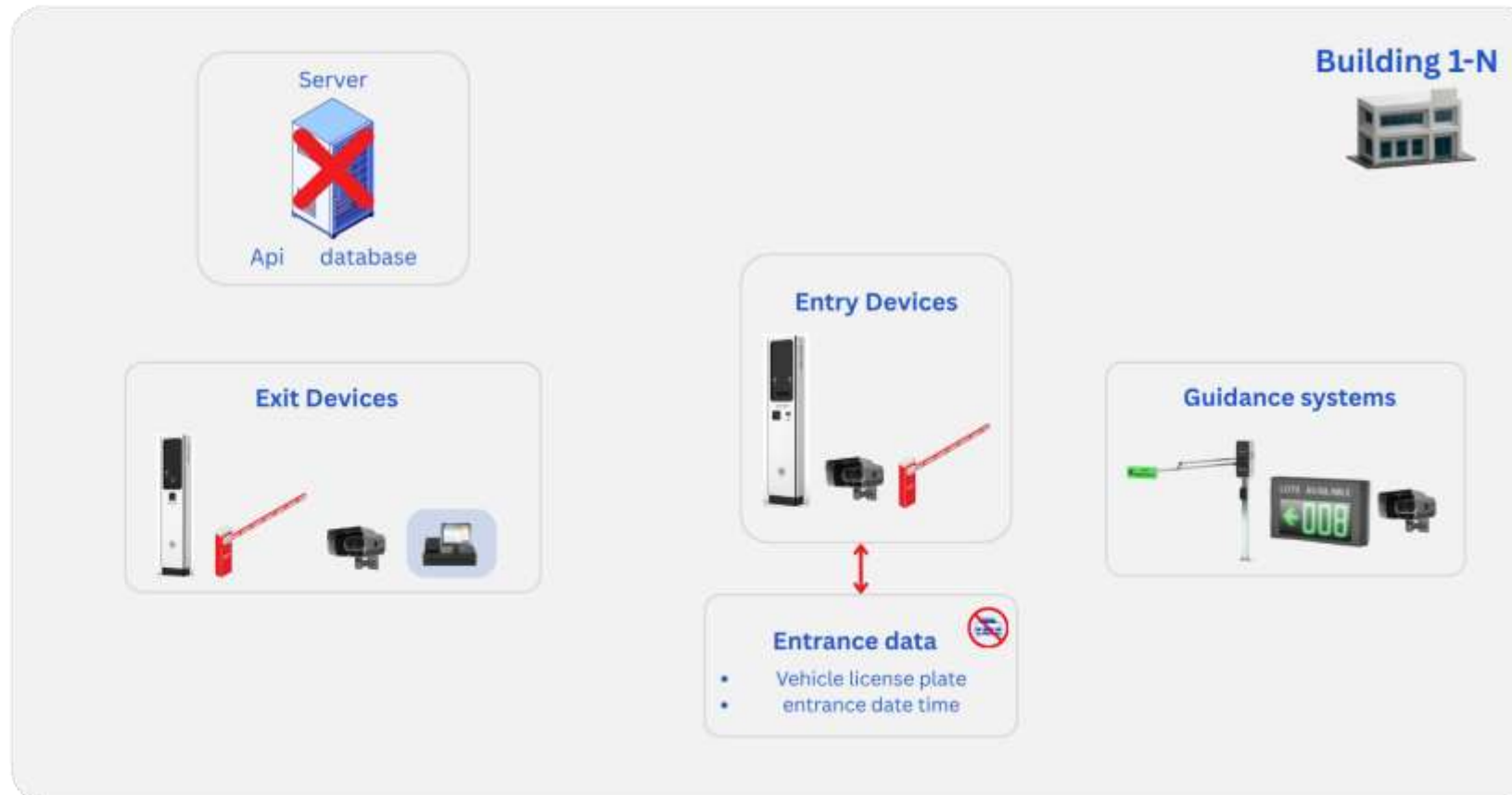
4. Complete Communication Loss POS Autonomous Mode:

- POS terminals operate independently
- Entry operations: Disabled (barriers remain closed for safety)
- Exit operations: Fully functional (revenue collection continues)
- Local fee calculation based on cached entry data
- Transactions queued and synchronized when connectivity restored
- Zero revenue loss on exit processing

Data Flow and Synchronization

- Exit Transactions → POS Local Storage → Building/Central Servers (upon recovery)

Complete communication loss “POS” autonomous mode



Scenario	Entry	Exit	Data integrity
Normal Operation	 Working	 Working	 Working
Normal Operation	 Working	 Working	 Working
Centralized Server Communication Loss	 Working	 Working	 Real-time sync
All Servers Down	 Disabled	 Working	 Working

Automatic Recovery

- *Building server restoration automatically detected*
- *Data synchronization from centralized to building servers*
- *POS terminals reconnect to building servers*
- *Full audit trail of all transactions maintained*
- *No manual intervention required*

3. Network Architecture

1. Connectivity Requirements Primary Connections:

- Building servers to centralized servers: Dedicated network link
- POS to building servers: Local area network (LAN)
- Hardware to building servers: RS485/TCP/IP Backup

Connections:

- POS to centralized servers: Direct failover link
- Centralized to hardware: Emergency control path

2. Security Features

- Encrypted communication (TLS)
- Network segmentation and VLANs
- Firewall protection
- Access control and authentication
- Comprehensive audit logging

4. Scalability

1. Growth Support

- Horizontal Scaling: Add unlimited building facilities
- Vertical Scaling: Increase capacity per building as needed
- Modular Design: Independent building operations
- Centralized Control: Single pane of glass management

2. Performance

- Sub-second transaction processing
- Support up to 5000 of parking spaces per building
- Real-time occupancy and guidance updates
- Concurrent multi-building operations

5. Business Benefits

1. High Availability

- 99.99% system uptime
- Multiple failover layers prevent service disruption
- Revenue collection never stops

2. Operational Continuity

- Exit operations guaranteed even during complete outages
- POS autonomous mode protects revenue
- Automatic recovery without staff intervention

3. Centralized Management

- Unified dashboard for all facilities
- Real-time reporting and analytics
- Remote monitoring and control
- Reduced operational overhead

4. Data Integrity

- Zero transaction loss guarantee
 - Automatic synchronization and reconciliation
 - Complete audit trails for compliance
-

Parking Guidance Management System

Guidance Management System

System Overview

The Ultrasonic Parking Guidance Management System (UPGS) is a real-time occupancy detection solution that guides drivers to available parking spaces efficiently. The system utilizes ceiling-mounted ultrasonic sensors to monitor individual parking spaces and integrates with digital signage, LED indicators, and the central Parking Management System to provide instant occupancy updates.

System Components

- Ultrasonic Sensors
 - Ceiling-mounted sensors above each parking space
- LED Indicators
 - Color-coded lights (Green = Available, Red = Occupied)
- Digital Signage
 - Zone-level displays showing available space counts
- License Plate Recognition (LPR) Camera
 - For reserved space verification
- Slot Locker Barriers
 - Physical access control for reserved spaces
- Central Management System
 - Real-time data processing and integration

Detection Methodology

Ultrasonic sensors emit sound waves and measure reflection time to determine space occupancy. When a vehicle is present, the shortened reflection distance triggers an "Occupied" status. Empty spaces are marked "Available" in real-time.

Guidance Management System

Operation Modes :

Mode 1: Standard Parking Spaces

Application: General parking areas without reservation capability

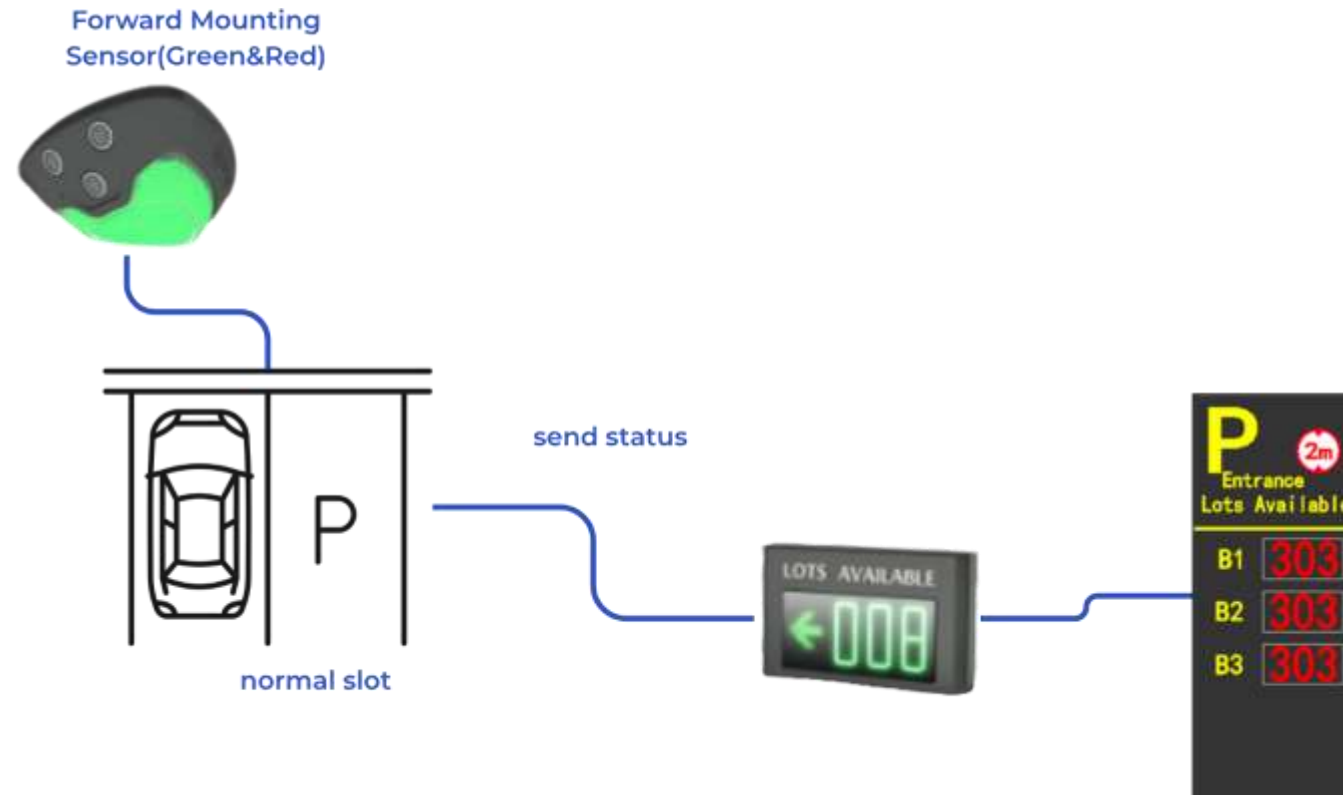
Operation:

- Ultrasonic sensor continuously monitors space occupancy
- Sensor transmits status to central system
 - LED indicator displays real-time status (Green → Available , Red → Occupied)
- Digital signage updates available space count per zone/level
- Mobile app displays real-time availability

Benefits:

Cost-effective, simple deployment, immediate visual guidance

Normal Parking



Guidance Management System

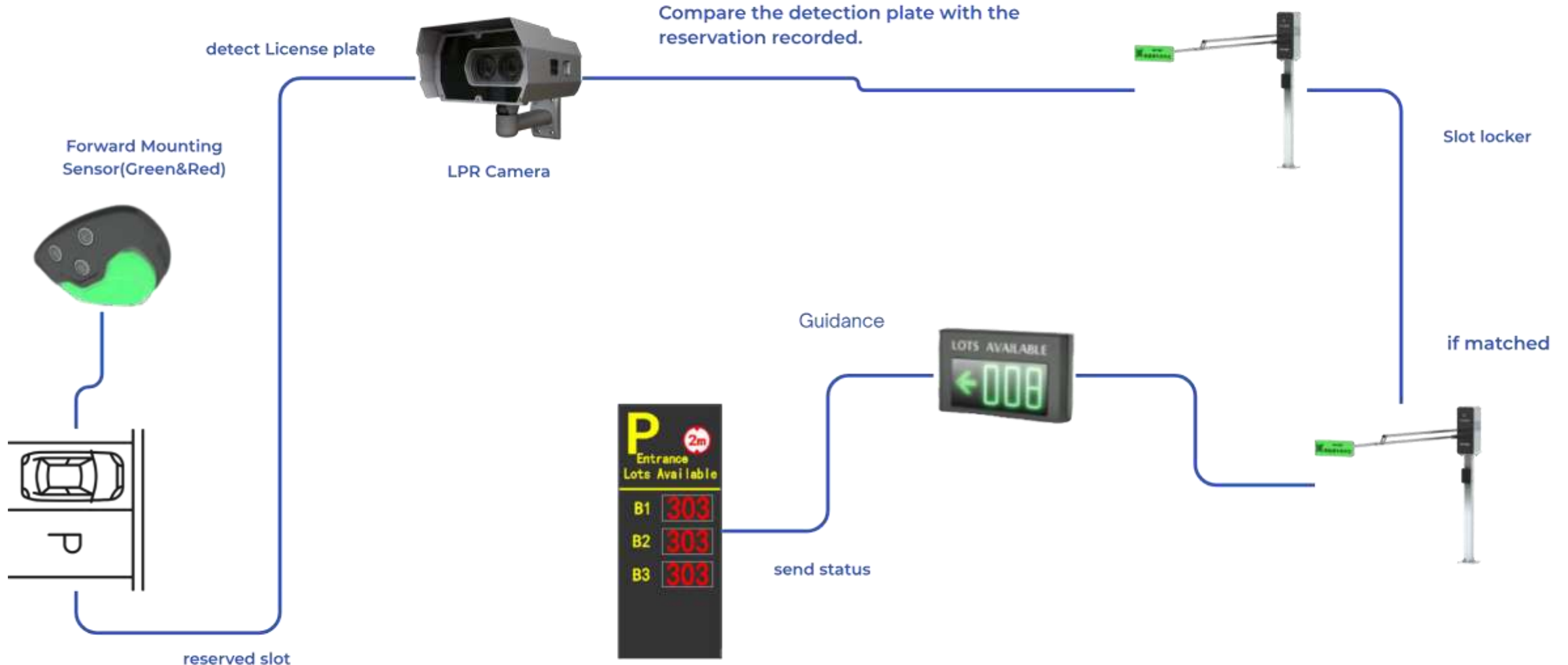
Mode 2: Reserved Parking Spaces

Application: Premium or subscription-based parking with access control

Additional Components:

- LPR Camera for license plate verification
- Slot Locker Barrier for physical access control [Operation:](#)
 - User reserves specific space via mobile app or subscription
 - System registers vehicle license plate and slot assignment
 - Upon approach, LPR camera scans vehicle license plate
 - System verifies plate against reservation database
 - Authorized: Slot locker opens automatically
 - Unauthorized: Locker remains closed; driver redirected
 - Ultrasonic sensor confirms vehicle presence
 - System updates occupancy status across all platforms

Reserved Parking



Smart Parking Management System

Smart Parking management system Overview

A fully integrated, smart parking management platform designed to deliver seamless experiences for residents, employees, and visitors across urban environments.

The solution combines a proven Parking Management System offering flexible deployment options that balance cost-effectiveness with cutting-edge innovation.

The system supports multiple entrance methods, including UHF subscription access, QR- based subscription entry, ticketless access via License Plate Recognition (LPR), and manual ticket issuance for visitors.

This platform ensures efficiency, automation, and exceptional user convenience, enabling smart, secure, and scalable parking operations



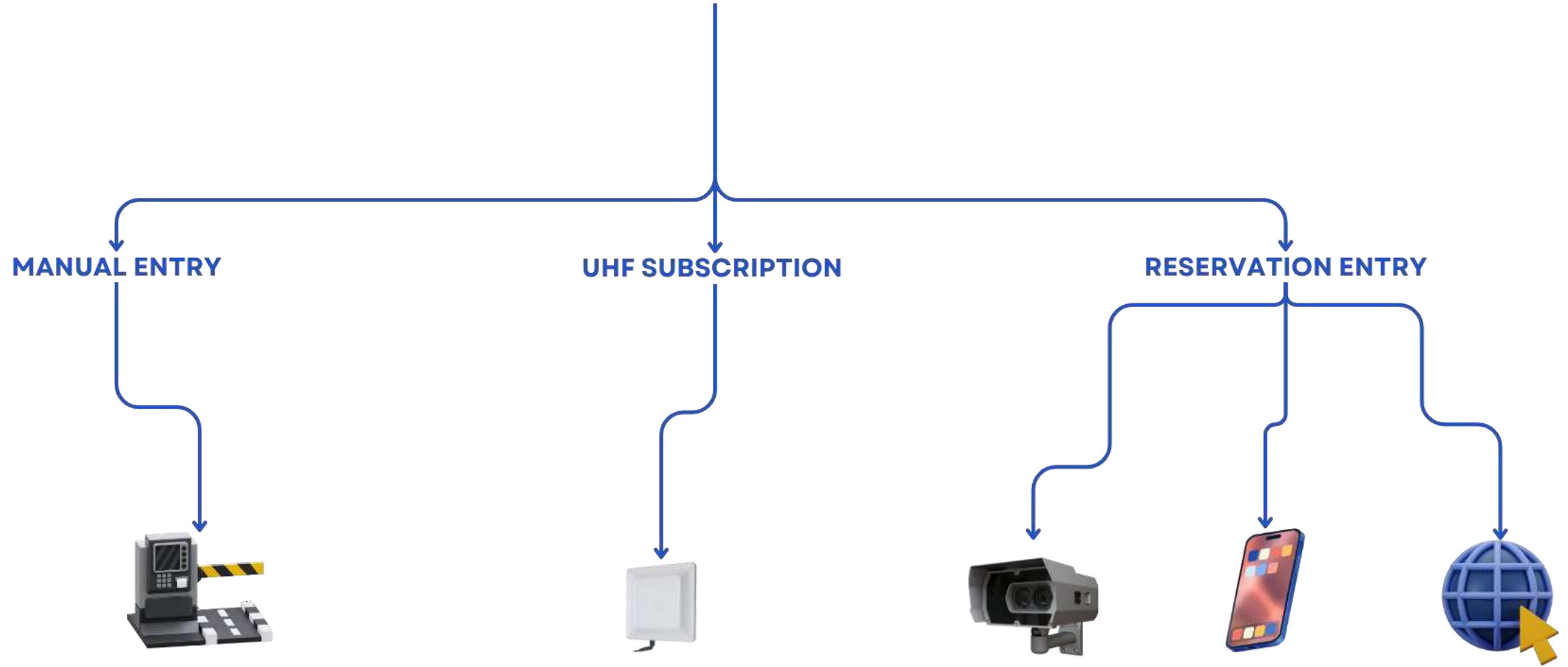
Entry/Exit Configuration



Devices Configuration		
Lanes	Devices	Quantity
Entrance	ANPR Camera	1
	Dispenser Ticket	1
	UHF Reader	1
	Automatic Barrier	1
Exit	ANPR Camera	1
	Automatic Barrier	1
	UHF Reader	1

ENTRANCE TYPES

ENTRANCE

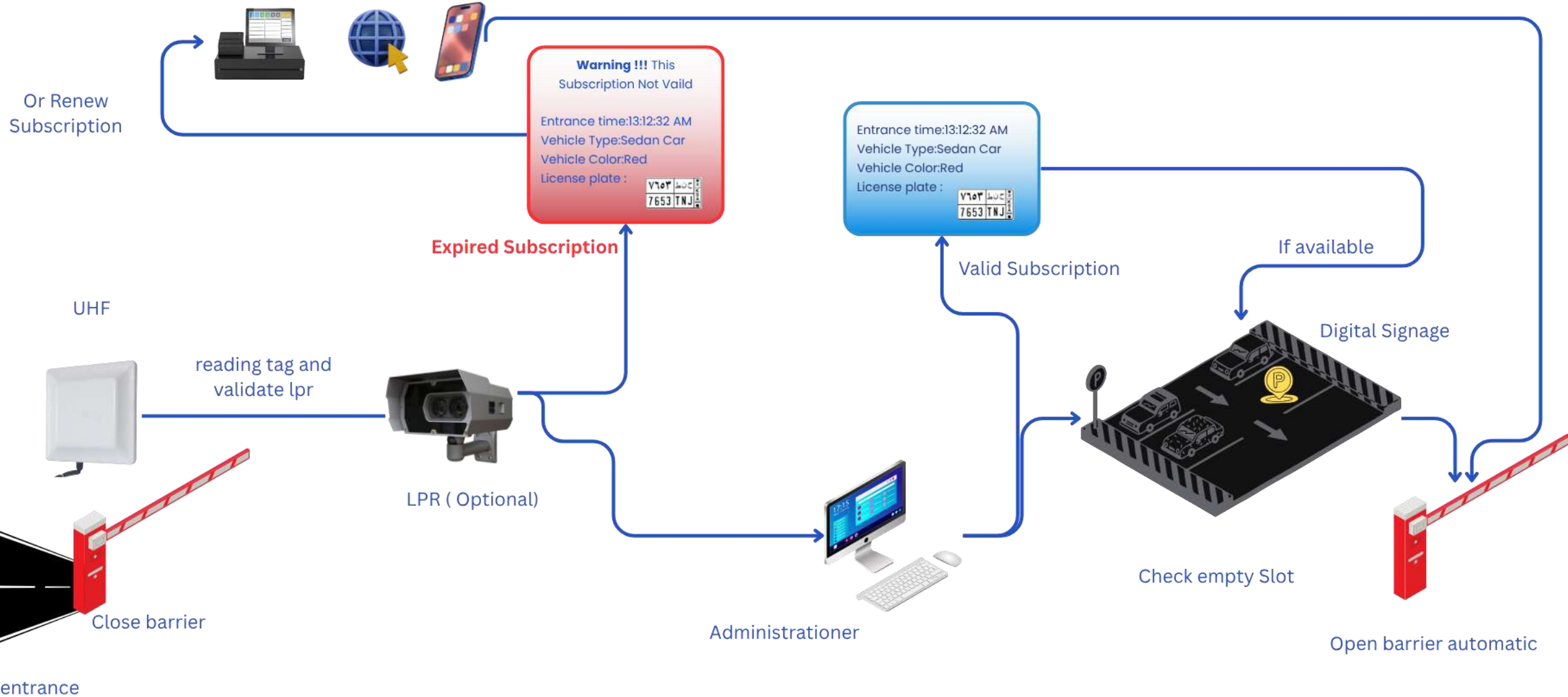


Entry Scenarios

1. UHF Subscription Entry

- The system validates the UHF tag.
- Upon successful validation:
- The system confirms that the subscriber already has a pre-reserved slot in their assigned main parking area.
- Gate opens automatically, and the operator is notified with subscriber details.
- Transaction data (LPR image, subscriber info, vehicle details, gate ID) is recorded.
- Digital signage guides the driver to their designated slot.
- Once parked, the slot LED indicator changes to red (occupied).
- For optional parking areas, the subscriber can enter if their subscription is valid and a free slot is available at that moment.

ENTRANCE SUBSCRIPTIONS (UHF)





Entry Scenarios

2. QR Code & LPR Subscription / Reservation Entry

- The user scans the QR code at the entrance.
- The system performs:
 - LPR recognition for vehicle identification.
 - QR-LPR data validation in the database.
 - Slot availability check.
- If validation succeeds:
 - Barrier opens, and the system logs the transaction data.
 - Digital signage guides the user to an available slot.
- The slot LED indicator turns red when occupied.

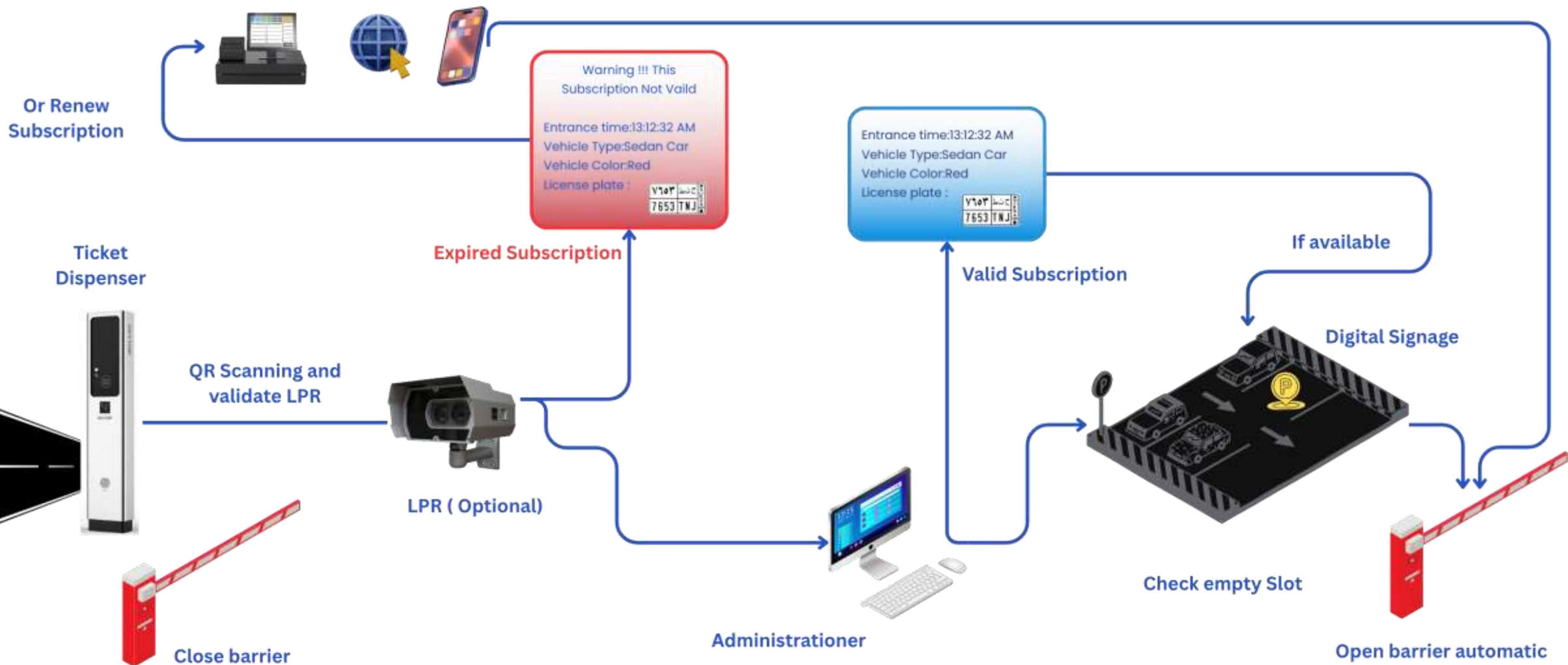
Error Handling:

Invalid QR Code: - expired or unrecognized

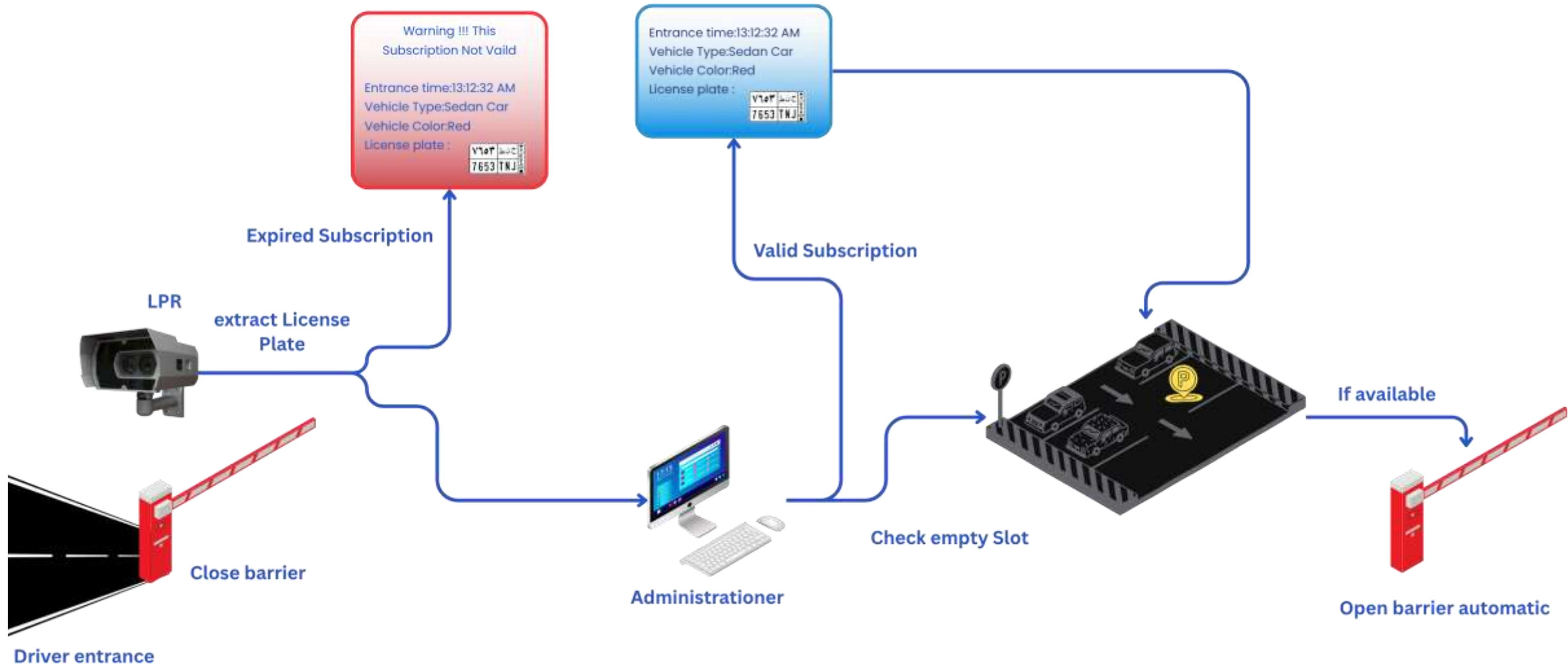
License Plate Mismatch: - vehicle doesn't match reservation

System prompts user to press manual entry button or contact assistance

ENTRANCE SUBSCRIPTIONS (QR CODE & LPR)



ENTACNE SUBSCRIPTIONS (LPR)



Entry Scenarios

3. Manual Entry

The user presses the ticket dispenser to issue a manual entry ticket.

The system captures the LPR data, opens the barrier, and records the entry event.

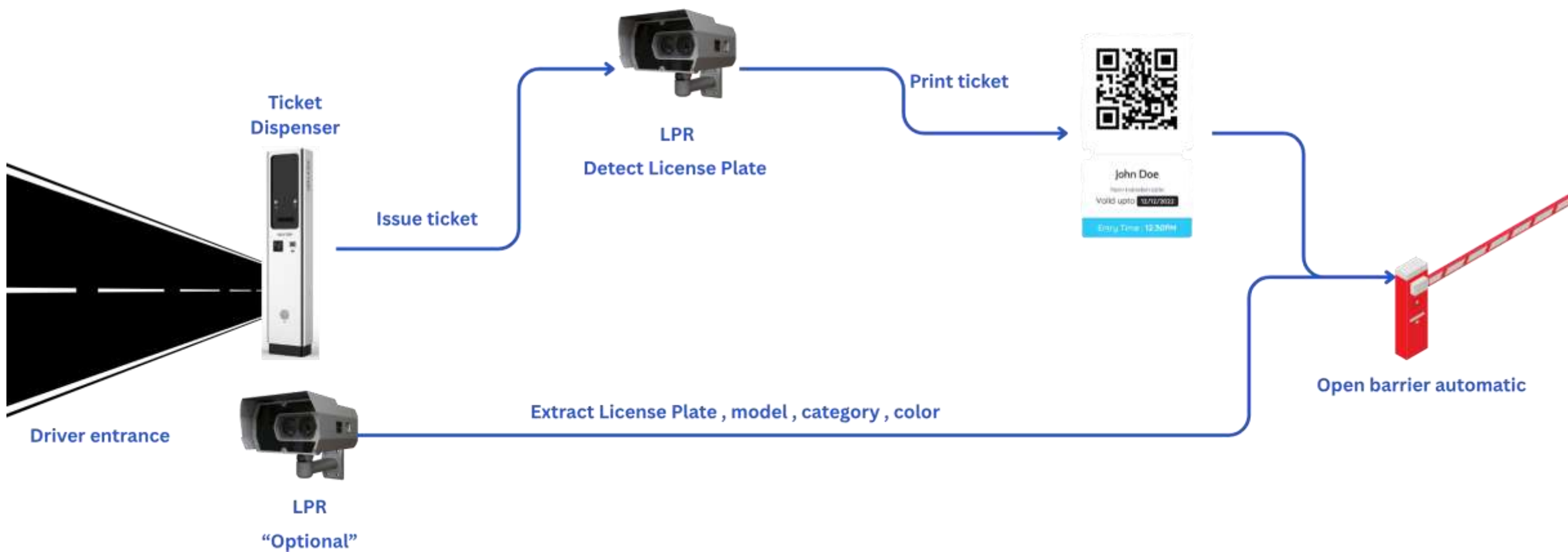
3. Ticketless Entry option Administrator Control:

The system includes configurable ticketless entry functionality that can be enabled or disabled from the central administration panel.

When Enabled:

- LPR camera captures license plate at entry
- No physical ticket printed
- Vehicle identified and tracked via license plate only
- Exit authorization based on LPR verification
- Payment kiosks accept license plate lookup for fee calculation [Use Cases:](#)
 - Registered user vehicles (automatic recognition)
 - Facilities preferring paperless operations
 - High-throughput entry lanes
- Environmental sustainability initiatives [Configuration:](#)
 - Toggle on/off per facility or system-wide
 - Backup manual ticketing remains available
 - Operator override capability maintained

MANUAL ENTRANCE (PUSH BUTTON)



Exit Scenarios

1. UHF Subscription Exit

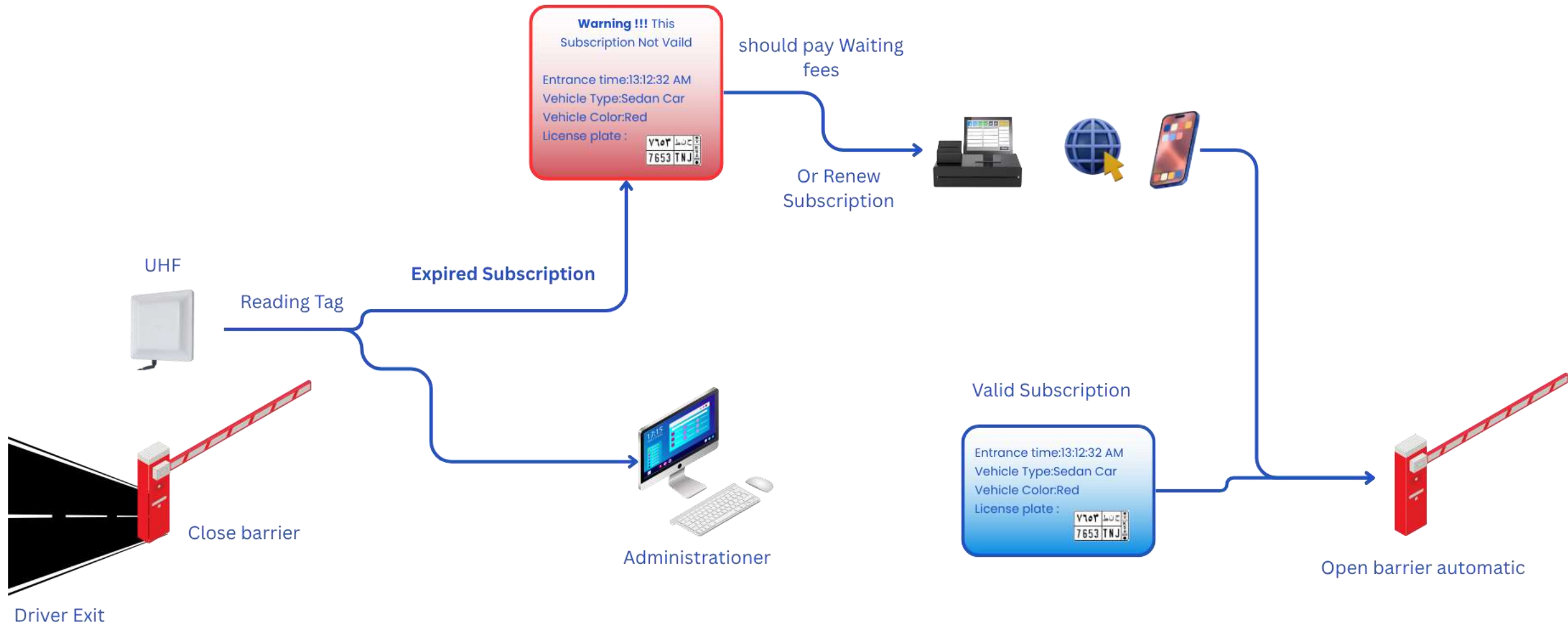
System validates exit authorization:

If valid:

- Barrier opens automatically
- Operators receive subscriber details.
- Parking duration logged for reporting
- LED indicator changes to green (available) Occupancy count updated in real-time
- The system records exit transaction data (LPR, subscriber info, gate ID).

EXIST (UHF)

Reserved Parking



Exit Scenarios

2. QR / LPR Subscription, Reservation

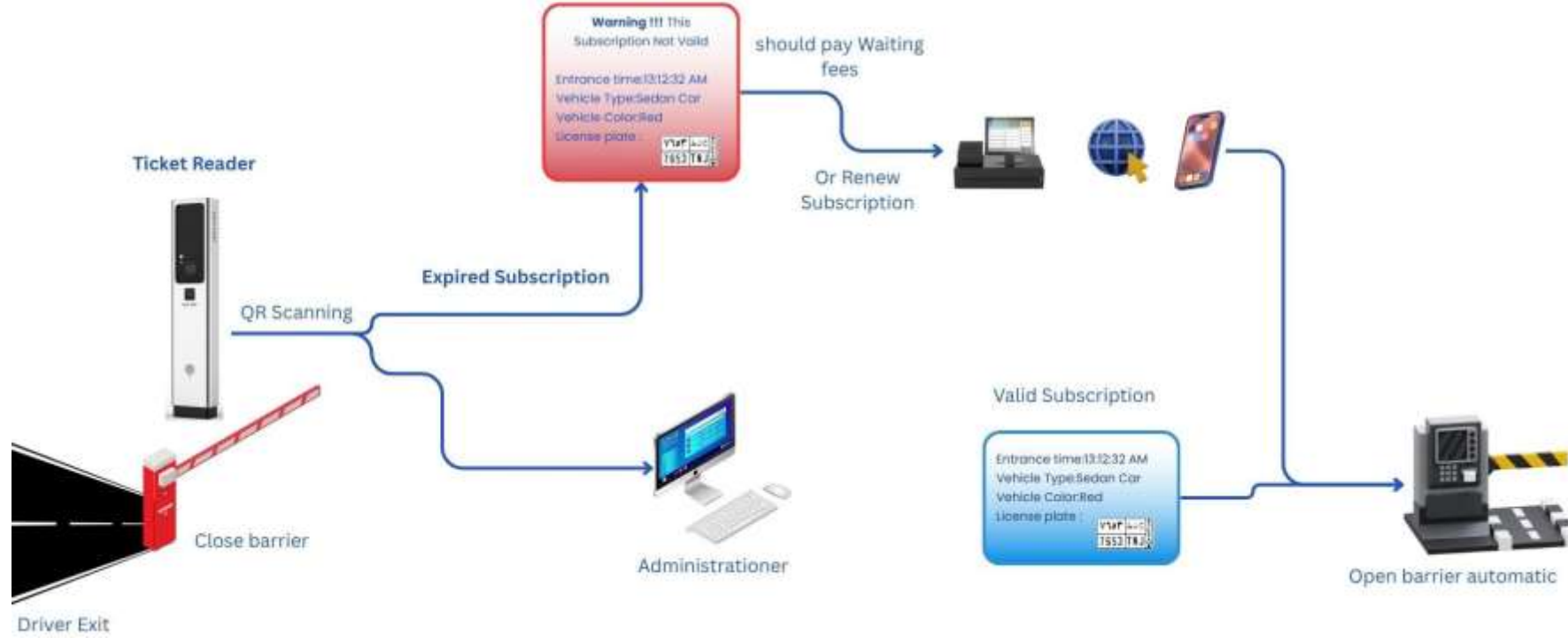
- Driver scans QR code at exit scanner.
- System validates subscription/reservation status
- For QR/LPR users Exit authorization validity.
- Barrier opens automatically upon authorization Parking session closed and slot released

Error Handling:

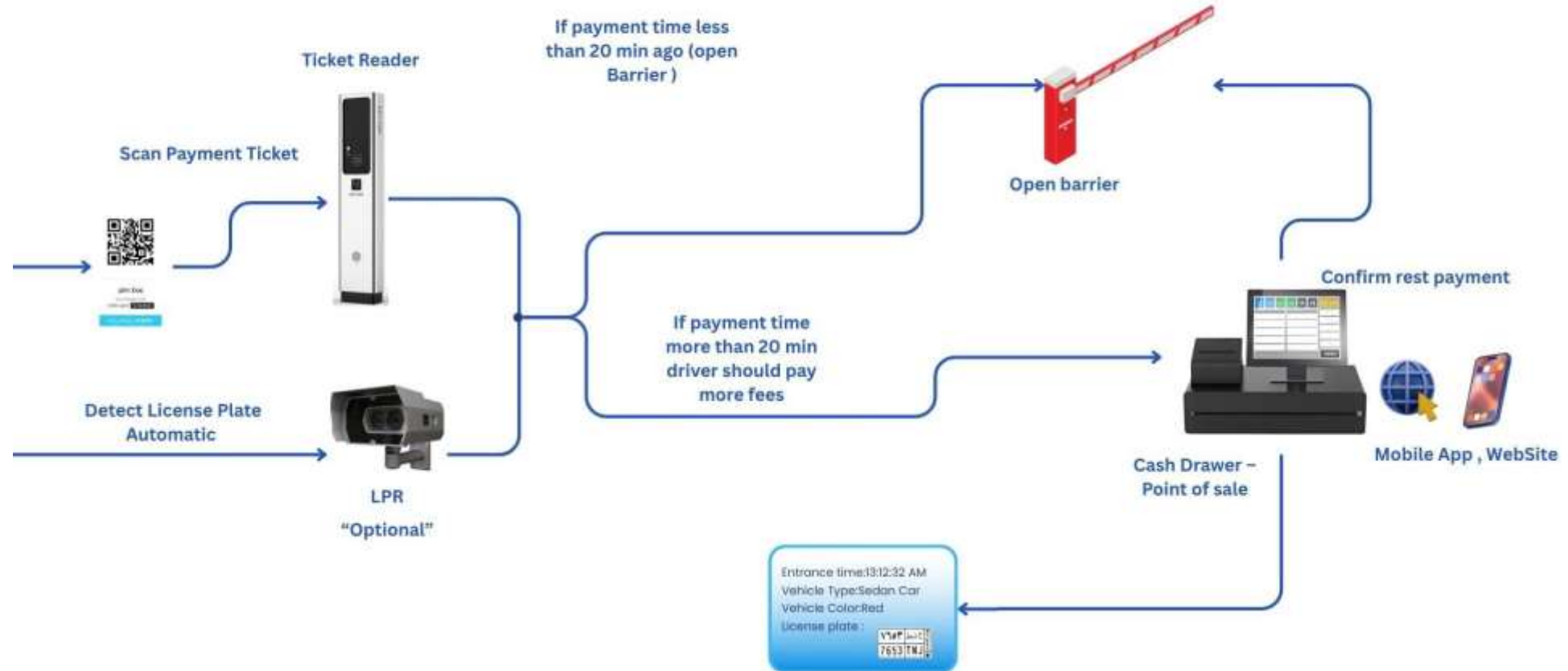
- "Invalid QR Code" - expired or unrecognized
- "License Plate Mismatch" - vehicle doesn't match reservation
- "Reservation Expired" - booking time window exceeded
- System prompts user to pay Extra Fee or contact assistance

Reserved Parking

EXIST (QR CODE)



EXIT(MANUAL)



Exit Scenarios

- The user can complete payment either at a KIOSK, POS cashier, or automatically via license plate recognition (LPR) for vehicles with registered plates.
- After payment, the user is granted a customizable 20-minute grace period to exit the parking facility.
- When approaching the exit barrier, the LPR camera automatically detects the vehicle's license plate, verifies payment status, and opens the barrier, eliminating the need for manual ticket scanning.

Supported payment methods include:

- Cash
- Visa
- Credit Card
- Digital Wallet
- Other integrated payment gateways

Budgetary Financial Proposal

Software

#	Description	Qty	Unit Price (SAR) without VAT	Total Price (SAR) without VAT
Smart Parking Guidance and Management Software				
1	Smart Parking Management Software (SPMS) Perpetual License	1	287,500.00	287,500.00
2	Smart Parking Guidance Software (SPGS) Perpetual License	1	201,250.00	201,250.00
3	SPMS and SPGS Site Server Perpetual License	10	11,500.00	115,000.00
4	Integration with ERP and Smart City Platform	2	172,500.00	345,000.00
5	5 Years Software Support	1	402,500.00	402,500.00
Total for SPMS and SPGS Software				1,351,250.00
Total for Software Excluding VAT (SAR)				1,351,250.00

Budgetary Financial Proposal

Hardware

#	Description	Qty	Unit Price (SAR) without VAT	Total Price (SAR) without VAT
Smart Parking Management Hardware				
1	POS Cashier	62	8,500.00	527,000.00
2	ANPR Camera (Entry/Exit)	93	15,000.00	1,395,000.00
3	UHF Reader	93	5,000.00	465,000.00
4	UHF Tag Programmer	5	5,000.00	25,000.00
5	UHF Tags	5000	20.00	100,000.00
6	Entrance / Exit Barrier	93	12,000.00	1,116,000.00
7	Entrance Ticket Dispenser	31	20,000.00	620,000.00
8	Exit Ticket Reader	62	21,000.00	1,302,000.00
9	Payment Kiosk	93	60,000.00	5,580,000.00
10	Thermal Paper (pack of 5000)	10	170.00	1,700.00
Total for Smart Parking Management Hardware				11,131,700.00
Smart Parking Guidance Hardware				
11	Ultrasonic Detector with LED Indicator	6000	350.00	2,100,000.00
12	Indoor One Way LED Display (Panel + Casing)	186	2,400.00	446,400.00
13	Indoor Two Way LED Display (Panel + Casing)	310	3,100.00	961,000.00
14	Main Entrance Display	62	28,500.00	1,767,000.00
15	Zone Control Unit (ZCU)	372	2,650.00	985,800.00
16	Accessories for installation (Excluding 1st and 2 nd fix)	1	800,000.00	800,000.00
Total for Smart Parking Guidance Hardware				7,060,200.00

Budgetary Financial Proposal

Hardware

#	Description	Qty	Unit Price (SAR) without VAT	Total Price (SAR) without VAT
Smart Parking Guidance and Management Common Hardware				
17	SPMS and SPGS Site Server	10	51,750.00	517,500.00
18	Smart Parking Centralized Management Server	7	74,750.00	523,250.00
19	Smart Parking Database Server	4	80,500.00	322,000.00
20	24x1G Ports PoE+ Switch with 2x10G Uplinks	35	7,000.00	245,000.00
21	8x1G Ports PoE+ Switch with 2x1G Uplinks	70	3,500.00	245,000.00
22	8x10G UTP, 8x10G Fiber Switch	70	15,000.00	1,050,000.00
23	10G SFP, Single Mode	70	500.00	35,000.00
24	1G SFP, Single Mode	140	200.00	28,000.00
25	Operator Workstation	5	7,000.00	35,000.00
Total for Smart Parking Guidance and Management Common Hardware				3,000,750.00
Total for Hardware Excluding VAT (SAR)				21,192,650.00

Budgetary Financial Proposal

Professional Services

#	Description	Qty	Unit Price (SAR) without VAT	Total Price (SAR) without VAT
1	Installation, Testing and Commissioning (Excluding Civil work, Main Power & Data work & Labor to connect PGS cabinets to central control room or to Client network)	1	2,500,000.00	2,500,000.00
2	4 Scheduled Preventive Maintenance Visits and 6 On-Call Visits Per Each Year (Off-site Team)	5	180,000.00	900,000.00
Total for Professional Services				3,400,000.00
Total for Professional Services Excluding VAT (SAR)				3,400,000.00

Budgetary Financial Proposal

Warranty Extension

#	Description	Qty	Unit Price (SAR) without VAT	Total Price (SAR) without VAT
1	Equipment warranty for additional for 3rd, 4th and 5th Years	1	2,806,100.00	2,806,100.00
Total for Warranty Extension				2,806,100.00
Total for Warranty Extension Excluding VAT (SAR)				2,806,100.00

Budgetary Financial Proposal

Optional Items

#	Description	Qty	Unit Price (SAR) without VAT	Total Price (SAR) without VAT
1	Special Park Slot ANPR Camera	1	3,343.23	3,343.23
2	Parking Lot Blocker Barrier	1	6,079.59	6,079.59

Scope of Work

Design, supply, installation, testing and commissioning of the proposed system, the scope of work includes the following:

- Prepare & submit engineering documents such as shop drawings and material submittal.
- Supply of system elements as per the above BOQ including delivery to the installation site.
- System elements installation (3rd fix) / one-time installation.
- Termination, configuration, testing, commissioning.
- On-Site training session for administrator and operator
- Closeout documents & system handover.

Exclusions

- All electrical works not limited to power outlets, UPS, cables, the connection to each machine and any other related site requirements.
 - All civil and MEP work not limited to foundation, excavation, containment & conduits, induction loop cutting & backfilling, protective bollards, and any other related site requirements.
 - Onsite heavy machinery for loading and unloading the machines upon delivery to storage area and installation locations.
 - Onsite convenient storage area to store electronics and sensitive materials
 - Control room arrangement and furniture.
 - Any insurance, NOC, government & authority NOC or approvals, taxes, or fees.
 - Credit/Debit card readers and bank contract agreement.
-

Terms & Conditions

Warranty:

- The proposed system will deliver with 2 years warranty from the date of the system handing over or 2 years and 2 months from the date of delivery whichever is earlier. Additional Warranty for 3rd, 4th and 5th years are offered as optional item.
- The warranty is applicable subject to warranty standard terms and conditions.

Prices:

- The offered prices are site delivery in Riyadh - KSA.
- The offered prices does not include VAT.
- The validity for prices is 30 Days from the date of this quotation.

Payment Terms:

- 60% of the total contract value as down payment.
- 30% of the total contract value upon material delivery (partial delivery is allowed).
- 10% of the total contract value upon installation and handover or maximum within 3 months from the material delivery date.
- Payments to be executed within 15 calendar days from invoice submission.

Special Note:

Note that this offer is based on best possible assumptions for 31 buildings according to current available information. The BoM might be changed once full information of all the buildings will be available.